



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
20-09-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-2

Time : 3 Hours

Maximum Marks : 240

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

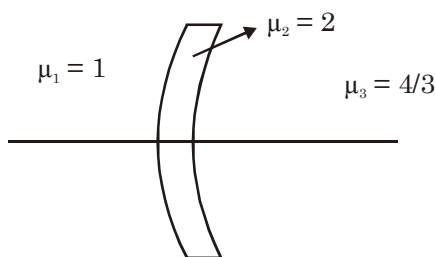
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

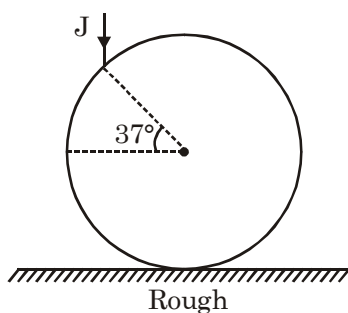
1. A thin lens of same radius of curvature 20cm for both surfaces, is having two different medium on its two sides extending upto infinity as shown in the figure. Then



- (A) It may behave as a converging lens of focal length 60cm.
- (B) It may behave as a diverging lens of focal length 60cm.
- (C) It may behave as a converging lens of focal length 80cm.
- (D) It may behave as a diverging lens of focal length 80cm.

Space for Rough Work

2. A sharp vertical impulse J is given to a uniform solid sphere of mass m and radius R placed on a rough horizontal surface. Choose the **CORRECT** options.



- (A) Impulse imparted to the sphere by the surface perpendicular to the surface will also have magnitude J .
 (B) Minimum value of coefficient of friction between sphere and surface to avoid slipping is

$\mu = \frac{4}{7}$ sphere rolls with backward slipping.

- (C) If $\mu = \frac{11}{20}$, sphere rolls with leftward slipping.

- (D) If $\mu = \frac{13}{20}$, sphere rolls with rightward slipping.

Space for Rough Work

3. If electric force (F) on a point charge 'q' due to another charge Q obeys following law

$$\vec{F} = \frac{Qq(1 - \sqrt{\alpha r})\vec{r}}{4\pi\epsilon_0 r^3}. \text{ Where } \alpha, \text{ positive constant, } r \text{ is position vector of charge } q' \text{ relative to } Q.$$

(A) Electric field due to point charge Q is $\vec{E} = \frac{Q(1 - \sqrt{\alpha r})}{4\pi\epsilon_0 r^3} \vec{r}.$

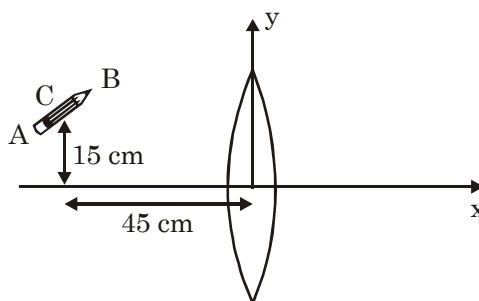
(B) $\oint \vec{E} \cdot d\vec{\ell}$ over a closed path will be equal to zero.

(C) Gauss's law $\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enclosed}}}{\epsilon_0}$ holds true.

(D) All statements are correct

Space for Rough Work

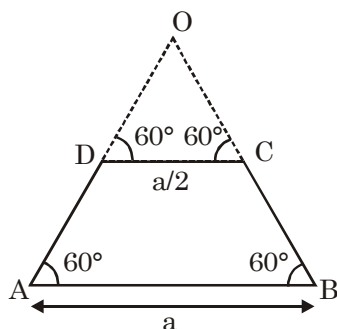
4. A $10\sqrt{2}$ cm long pencil is placed at an angle of 45° with optical axis. Its centre is 15 cm above the optics axis and 45.0 cm from a lens with 20.0 cm focal length as shown in the figure. Assume that the diameter of the lens is large enough for the paraxial approximation to be valid. For approximate position of image choose the **CORRECT** option(s) :



- (A) The location of image of point A is (30 cm, -6.10 cm)
 (B) The location of image of point B is (40.0 cm, -20.0 cm)
 (C) The location of image of point C is (36.0 cm, -12.0 cm)
 (D) The location of image of point A is (33.3 cm, -6.6 cm)

Space for Rough Work

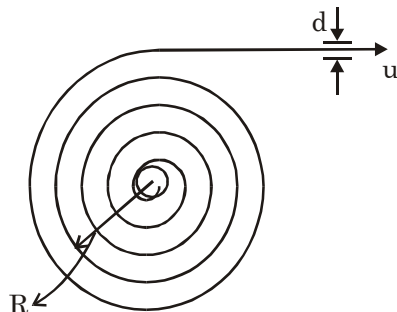
5. Consider a uniformly charged sheet ABCD, which is a part of an equilateral triangular sheet of a side a as shown in the figure. Choose the correct options regarding the electric field E at point O due to this sheet.



- (A) Magnitude of $\vec{E}$, increases with the increase in a (Keeping charge density same)
- (B) Magnitude of E , decreases with increase in a (keeping total charge same)
- (C) If charge density is σ and $a = 1$ m, magnitude of E is equal to $\frac{7\sigma}{44\epsilon_0} \ln \sqrt{2}$.
- (D) If charge density is σ and $a = 2$ m, magnitude of E is equal $\frac{7\sigma}{22\epsilon_0} \ln \sqrt{2}$.

Space for Rough Work

6. A disc is free to rotate about an axis passing through its centre and perpendicular to its plane. The moment of inertia of the disc about its rotation axis is I . A light ribbon is tightly wrapped over it in multiple layers. The end of the ribbon is pulled out at a constant speed of u . Let the radius of the ribboned disc be R at any time and thickness of the ribbon be d ($d \ll R$).

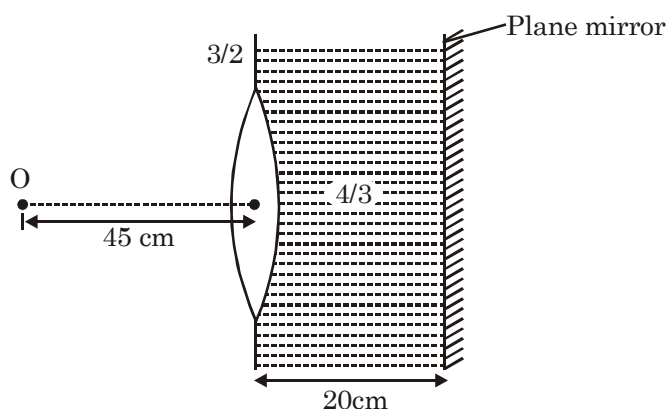


- (A) Angular acceleration of the disc at any instant is proportional to $\frac{1}{R^3}$, where R is radius of the ribboned disc.
- (B) Angular acceleration of the disc at any instant is proportional to $\frac{1}{R^2}$, where R is the radius of the ribboned disc.
- (C) Force required to pull the ribbon at constant speed u is $F = \frac{lu^2d}{2\pi R^4}$.
- (D) Force required to pull the ribbon at constant speed u is $F = \frac{lu^2d}{\pi R^4}$.

Space for Rough Work

7. Electric field in a region is given as $\vec{E} = (10 - 5x)\hat{i}$. A charge particle of mass 5kg and charge $Q (=1\text{C})$ is situated at origin and free to move in given electric field. Then choose the **CORRECT** options (Neglect gravity) :
- (A) Motion of charge particle is Oscillatory.
 (B) Maximum displacement of charge particle from origin is 4 SI units.
 (C) Maximum velocity gain by charge particle is 2 SI units.
 (D) The position of charge particle, when velocity gained by particle is maximum, is 2 SI units.

8. Figure shows an object 'O' placed in front of a biconvex lens ($\mu = \frac{3}{2}$) behind which there is water ($\mu = \frac{4}{3}$) and a plane mirror at a distance of 20m. A point object O is placed at a distance of 45 cm from the lens as shown in the figure. The position of the image of the object coincides with the object itself, then radius of curvature of the lens is.



- (A) 20cm (B) 45cm (C) 60cm (D) 30cm

Space for Rough Work

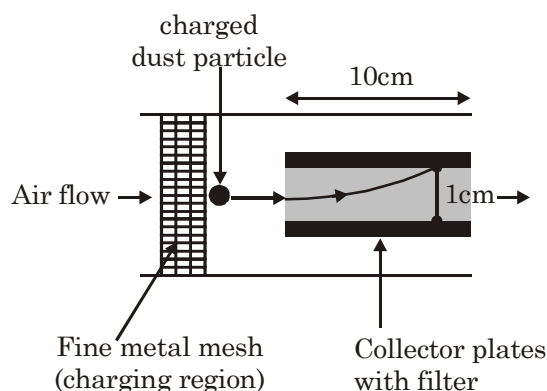
SECTION-I(ii) : (Maximum Marks : 16)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
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Space for Rough Work

Paragraph for Questions 9 and 10

An air ionizer filters particles of dust, pollen, and other allergens from the air using electric. In one type of ionizer (see diagram), a stream of air is drawn in with a speed of 3.0 m/s . The air passes through a line, highly charged wire mesh that transfers electric charge to the particles. Then the air passes through parallel collector plates that attract the charged particles and trap them in a filter. Consider a dust particle of radius $6.0 \mu\text{m}$, mass $1.0 \times 10^{-13} \text{ kg}$, and charge $100e$. The plates are 10 cm long and are separated by a distance of 1.0 cm , ignore gravity. Assume that the speed of dust particles does not change after crossing the mesh.

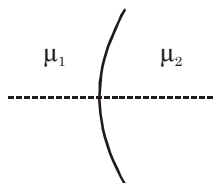


9. Ignoring drag force, what would be the minimum potential difference between the plates to ensure that all the particles get trapped by the filter?
 (A) 2250 V (B) 3750 V (C) 1500 V (D) 1125 V
10. Consider a particle injected in the lowest point of the plates at the above potential difference. Ignoring air drag, at what speed would the particle be moving relative to the stream of air just before hitting the filter?
 (A) 0.6 m/s (B) 3.6 m/s (C) 3 m/s approx (D) 6 m/s approx.

Space for Rough Work

Paragraph for Questions 11 and 12

Consider the curved surface separating two media, μ_1 and μ_2 as shown in figure below. Light is incident on the curved surface from left



11. If $\mu_1 = 1.5$, $\mu_2 = 1$. Choose the correct statement(s)
- (A) Real object real image combination is not possible.
 - (B) Virtual object real image combination is possible.
 - (C) Virtual object virtual image combination is not possible.
 - (D) Real object virtual image, combination is possible.
12. If $\mu_1 = 1$, $\mu_2 = 1.5$. Choose the correct statement(s)
- (A) Real object real image combination is not possible.
 - (B) Virtual object real image combination is possible.
 - (C) Virtual object virtual image combination is not possible.
 - (D) Real object virtual image, combination is possible.

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

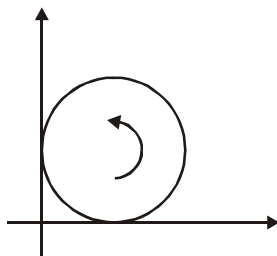
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

										+	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

										-	
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
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9	9	9	9	9	9	9	9	9	9	9	9

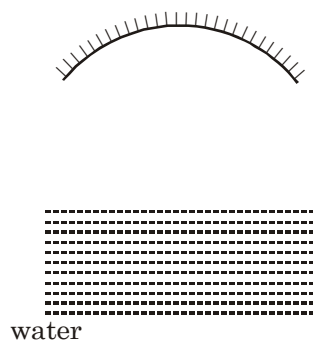
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A uniform cylinder of radius R ($= 3\text{m}$) is spin about its axis at an angular velocity ω_0 ($= 4\sqrt{10\pi}$ rad/sec) and placed between two perpendicular wall. The coefficient of friction between the walls and cylinder is μ ($= 2$). Then, how many turns will the cylinder make before it stops.

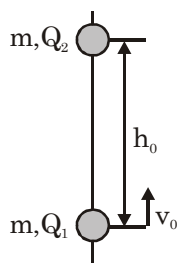


Space for Rough Work

2. A concave mirror of focal length 15 cm is placed in air at a height of 45 cm above the water surface as shown. The axis of the mirror is vertical and the reflecting surface faces the water surface. A point object is placed on the axis of concave mirror at a distance 35 cm above the water surface. Find the position of image formed after two successive processes- first reflection from the concave mirror and then first refraction from the water (refractive index of water $\frac{4}{3}$). The distance of image from water surface (in m) is



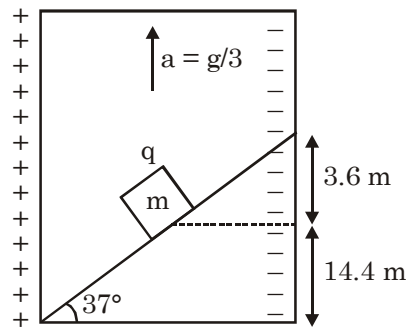
3. At the lower end of a vertically positioned insulator bar shown in the figure there is a pearl of the mass $m = 10^{-4}$ kg and a charge of Q_1 . Above it at a height $h_0 = 20$ cm, there is another pearl with the same mass m and electric charge Q_2 resting in equilibrium. At a given moment we kick the lower pearl and it starts upwards at a velocity of $v_0 = 2$ m/s. How close (in cm) can the lower pearl get to the upper one at most ? (The pearls can move along the bar without friction.)



Space for Rough Work

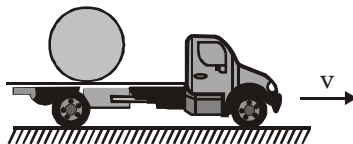
4. A point object is placed at a distance 25 cm from a convex lens of focal length 22cm. If a glass slab of refractive index 1.5 is inserted between the lens and object, then the image is formed at infinity. Find the thickness of glass slab (in cm).
5. A small block of mass m is kept on the smooth inclined plane of angle 37° , placed in an elevator going upward with acceleration $a = g/3$, as shown in the figure. A horizontal electric field E perpendicular to the left and right vertical wall of the elevator exists. The charge on the block is $+q$. Find the time (in sec) when the block collides with the vertical wall,

(take $g = 10 \text{ m/s}^2$ and $\frac{qE}{m} = 25 \text{ m/s}^2$)

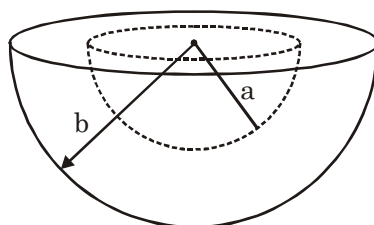


Space for Rough Work

6. A solid cylinder rolls from the back of a large truck travelling at 10 m/s to the right. The cylinder is travelling horizontally at 8 m/s to the left relative to an observer in the truck. The ball lands on the roadway 1.25 m below its starting level. How far behind the truck does it land (in m)?



7. A uniformly charged hemisphere of radius b and charge density ρ has a hemispherical cavity of radius a ($a = \frac{b}{2}$) cut from its centre. If the potential at the centre of the cavity is $\frac{\rho b^2}{16\epsilon_0}$ then find 'n'.



8. A man of height = $\frac{3}{2}$ m, wants to see himself in plane mirror from top to bottom. The plane mirror is inclined with vertical wall at angle $\alpha = 53^\circ$. If the least size of mirror to see him is x m, the distance of eye from mirror is $d = 3$ m, find the value of x .

Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 32)**

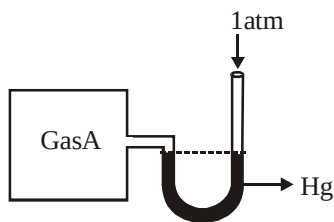
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-
1. 1 mole of parent 'ic' acid – $\text{H}_2\text{O} \longrightarrow (\text{A})$
If solid (A) has 3 allotropes then which of the following statement(s) is/are **INCORRECT** regarding one of the allotropes of (A) which exist at room temperature?
(A) It is a planar structure
(B) It has X-O-X linkage (X = central atom)
(C) It is a meta acid
(D) All atoms are having same hybridisation
2. Select **CORRECT** order between given compounds.
(A) $\text{H}_2\text{O} > \text{H}_2\text{O}_2$: Strength of hydrogen bonding
(B) $\text{H}_2\text{O} > \text{H}_2\text{O}_2$: Boiling point
(C) $\text{P}_4\text{O}_{10} > \text{P}_4\text{O}_6$: Number of P-O-P linkage
(D) Glycerol > Ethylene glycol : Viscosity

Space for Rough Work

3. When a gaseous mixture of C_2H_6 and C_3H_8 gases is burnt completely in excess of O_2 gas, 8.8 gm CO_2 and 5.04 gm H_2O vapours are formed. The correct information(s) related with the initial mixture is /are
- (A) Equal moles of both the gases were taken.
 - (B) Equal masses of both the gases were taken.
 - (C) Average molar mass was 37 gm/mol.
 - (D) The total mass of oxygen used in combustion was 10.88 gm.
4. Consider the following statements about Maxwell's distribution of velocities. The correct statement(s) is/are -
- (A) As temperature increases, the peak (maxima) of the curve shifts away from origin for a fixed mass of gas.
 - (B) As temperature increases, the most probable velocity increases, but the fraction of gas molecules travelling at most probable velocity decreases
 - (C) The area under the curve is independent of temperature
 - (D) As (T/M) remains constant for different samples of gases, the shape of curve remains same.

Space for Rough Work

5. An open ended mercury manometer is used to measure the pressure exerted by a trapped gas as shown below. Initially there is no difference in mercury levels in both the arms - As A dissociates upon sparking as follows : $A(g) \rightarrow B(g) + 3C(g)$

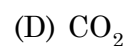
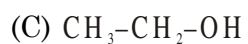
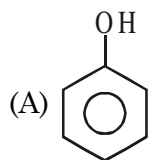
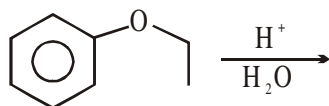


if pressure of gas A decreases to 0.9 atm, then which of the following are true at that instance at 300 K

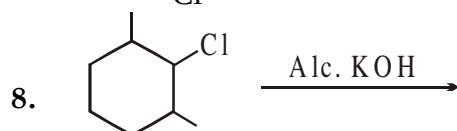
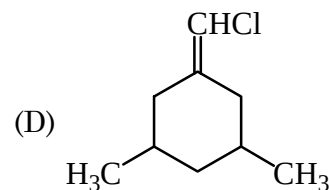
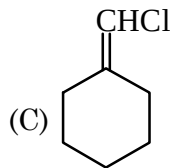
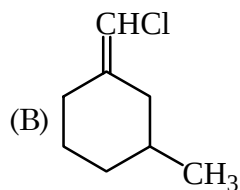
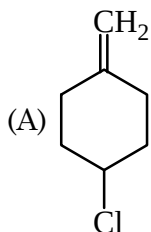
- (A) Total pressure of gaseous mixture = 1.3 atm
- (B) Increase in total pressure = 0.3 atm
- (C) Difference in mercury level is = 98.8 mm
- (D) Difference in mercury level is = 228 mm

Space for Rough Work

6. Major product/s of reaction is/are



7. Among the following compounds, geometrical isomerism exhibited by :



Correct statement regarding reaction.

(A) Product can show geometrical isomerism

(B) It is a example of E_2 mechanism

(C) $(\pm)$ 1,3-Dimethyl cyclohexene is obtained as a product

(D) Racemic mixture of alcohols are obtained as product

Space for Rough Work

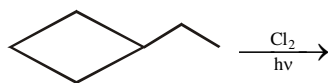
SECTION-I(ii) : (Maximum Marks : 16)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 9 and 10

Radicals are obtained by homolysis of bond. Since energy is required to break a bond in homolytic pattern, hence light is a possible energy source and ultraviolet light with an associated energy of 586 KJ/mole ($h\nu$) is used for homolysis. It can decompose many organic compounds including DNA in skin cells.

During halogenation in presence of ultraviolet light a free radical substitution reaction takes place. A reaction is given as—



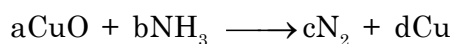
- Total number of possible monochloride obtained.
 (A) 5 (B) 9 (C) 10 (D) 6
- Total number of fractions obtained on fractional distillation of product mixture
 (A) 7 (B) 5 (C) 8 (D) 6

Space for Rough Work

Paragraph for Questions 11 and 12

Chemical reactions in which substance undergo oxidation and reduction are called redox reaction. In redox reaction, reactants react with each other such that number of electrons involved in the oxidation and reduction process must be same. In oxidation process oxidation number of atoms increases due to loss of electron and in reduction process oxidation number decreases due to gain of electrons.

For the given redox reaction



11. The values of a & b respectively are:

- (A) 2, 3 (B) 3, 2 (C) 4, 3 (D) 3, 4

12. Number of electrons involved per molecule of N_2 is-

- (A) 3 (B) 2 (C) 4 (D) 6

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

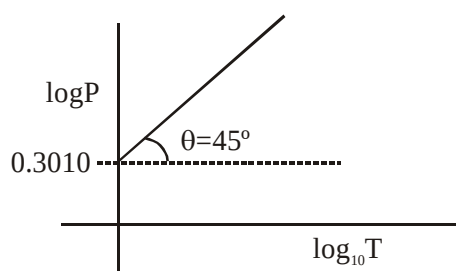
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

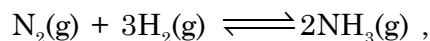
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. In a rigid container of volume 82.1 ml, CH_4 gas present. Calculate number of moles of CH_4 gas present, if graph between $\log P(\text{atm})$ vs $\log T(\text{K})$ is obtained as follows.



Space for Rough Work

2. 1 mol of nitrogen is mixed with 3 moles of hydrogen in a $\sqrt{3}$ litre container where 66.67 % of nitrogen is converted in to ammonia by the following reaction :

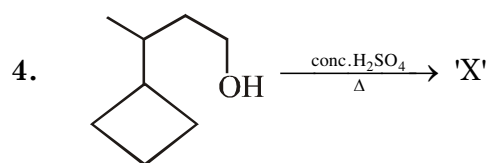


then the value of K_c for the reaction $\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2$ will be -

3. $\Psi_{1s} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} e^{-\frac{r}{a_0}}$ (for H-atom)

Find the ratio of probability density at $r = a_0$ to the probability density at nucleus ($r = 0$).

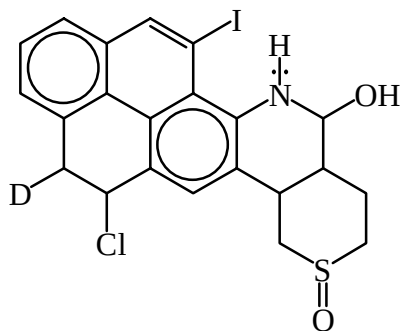
[Given : $\ln\left(\frac{1}{0.14}\right) = 2$]



Total number of 1,2 shifts (rearrangements) involve in the above reaction.

Space for Rough Work

5. How many chiral center are present in this compound



6. An unknown unbranched hydrocarbon with molecular formula C_5H_{10} decolourise $Br_2 + H_2O$ and on reaction with $H-Cl/CCl_4$ give 2-chloro pentane as major product. Identify total number of possible reactants (excluding stereochemistry) -

Space for Rough Work

7. An element X of group 7 has ratio of filled orbitals to vacant orbitals upto outer most shell is 1. The magnetic moment of its dipositive ion is _____.
8. Find the sum of the axial d-orbitals which are used in the hybridisation of central atom(s) of the given species :
- (i) XeF_6 (ii) PF_2Cl_3 (iii) SO_4^{2-} (iv) TeF_6^{2-}
(v) POCl_3

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

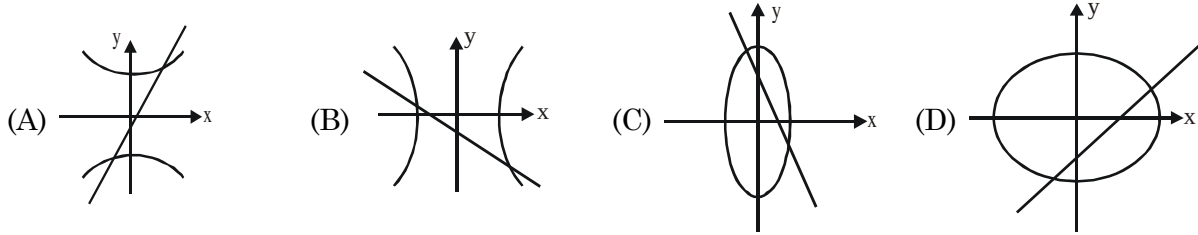
1. Let $Q(n)$ denotes the number of seven digit numbers divisible by 9 which can be formed by using 7 distinct digits out of 1,2,3,4,5,6,7,8,9. Then which of the following is/are not the value of $Q(n)$.
 (A) $\underline{9}$ (B) $2.\underline{7}$ (C) $4.\underline{7}$ (D) $3.\underline{7}$
2. The coefficient of x^{324} in the expansion of $(x+1)(x-2)^2(x+3)^3(x-4)^4 \dots (x-24)^{24}(x+25)^{25}$ is-
 (A) 325 (B) $\left(\sum_{k=0}^8 {}^8C_k + 4 \cdot \sum_{s=0}^4 {}^4C_s + \sum_{r=0}^2 {}^2C_r + {}^5C_0 \right)$
 (C) $1^2 - 2^2 + 3^2 - 4^2 + \dots - 24^2 + 25^2$ (D) $1^1 + 2^2 + 3^3 + 4^4 + \dots + 25^{25}$

Space for Rough Work

3. If the expression $(\cos 3\theta + \sin 3\theta) + (2\sin 2\theta - 3) \cdot (\sin \theta - \cos \theta)$ is positive, then θ belongs to -
- (A) $\left(2n\pi - \frac{3\pi}{4}, 2n\pi + \frac{\pi}{4}\right), n \in \mathbb{I}$ (B) $\left(2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{6}\right), n \in \mathbb{I}$
- (C) $\left(2n\pi - \frac{\pi}{3}, 2n\pi + \frac{\pi}{3}\right), n \in \mathbb{I}$ (D) $\left(2n\pi - \frac{\pi}{4}, 2n\pi + \frac{3\pi}{4}\right), n \in \mathbb{I}$
4. In a triangle ABC, if $\sin^4 A + \sin^4 B + \sin^4 C = 2\sin^2 A \sin^2 B + \sin^2 B \sin^2 C + 2\sin^2 C \sin^2 A$, then angle A can be equal to -
- (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{5\pi}{6}$ (D) $\frac{7\pi}{6}$
5. Given $x = \log_2 \left(1 + \frac{1}{2} \sum_{k=1}^{11} {}^{12}C_k\right)$, $\log_y 3 = 4$ and $\log_{y^2} 27 = \frac{3z}{2}$ then which of the following is/are true-
- (A) $z^2 - y^4 = 13$ (B) $x = 11$
- (C) $x^2 - y^4 + z^2 = 134$ (D) $x + z^2 \div y^4 = 9$

Space for Rough Work

6. Suppose $a, b \in \mathbb{R}$ where $ab \neq 0$, then the graph of the straight line $ax - y + b = 0$ and the conic section $bx^2 - ay^2 = ab$ can be -



7. If equations $ax^2 - (a + b)x + b = 0$ & $bx^2 + (b - c)x - c = 0$ has exactly one root in common $\{a, b, c \neq 0\}$, then which of the following can be correct -
- (A) $b^2 = ac$ (B) $-a = b \neq c$ (C) $b = a \neq c$ (D) $-a \neq b = c$
8. If x_1 & x_2 are any two positive values of x in $[0, 2\pi]$ for which $|\cos x|$, $2\cos x$, $(4\sin^2 x - 1)$ are in G.P., then $|x_1 - x_2|$ can be equal to
- (A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{3}$ (D) π

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 16)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
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Paragraph for Questions 9 and 10

An enemy's jeep (treat it as point object) starts from a point (6, -2) along the path described by the curve $y^2 = 4x - 20$ such that its ordinate increases. A point soldier is located at origin notices that at a point P on the parabolic path the jeep is closest to him and a point Q is on the parabolic path such that OQ is tangent to the parabola (where O is origin).

On the basis of above information, answer the following questions :

9. Co-ordinates of Q are -

(A) $(10, 2\sqrt{5})$ (B) (6, 2) (C) $(7, 2\sqrt{2})$ (D) $(8, 2\sqrt{3})$

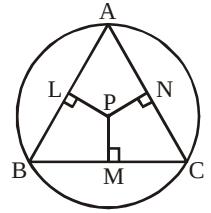
10. Area of ΔPOQ is-

(A) $\sqrt{5}$ (B) $\sqrt{10}$ (C) $5\sqrt{5}$ (D) 10

Space for Rough Work

Paragraph for Questions 11 and 12

In $\triangle ABC$, if P is circumcentre & let L, M, N are the foot of perpendiculars drawn from P to the sides AB, BC & CA respectively as shown in figure.



On the basis of above information, answer the following questions :

11. If areas of smaller sectors APB, BPC, CPA are denoted by $\Delta_1, \Delta_2, \Delta_3$ & $\Delta_1 : \Delta_2 : \Delta_3 = 2 : 3 : 4$, then $PL : PM : PN$ is equal to -
- (A) $\sin 40^\circ : \sin 60^\circ : \sin 80^\circ$ (B) $\cos 40^\circ : \cos 60^\circ : \cos 80^\circ$
 (C) $\tan 40^\circ : \tan 60^\circ : \tan 80^\circ$ (D) $\sin 10^\circ : \sin 30^\circ : \sin 40^\circ$
12. The area of $\triangle LMN$ is equal to (where Δ represents the area of triangle ABC) -
- (A) $\frac{\Delta}{2}$ (B) $\frac{\Delta}{4}$ (C) $\frac{\Delta}{3}$ (D) $\frac{\Delta}{16}$

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
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4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If the line $y = \sqrt{2}x - 4k\sqrt{2}$ is normal chord of parabola $y^2 = 2x$ whose length is $a\sqrt{a}$ ($a \in \mathbb{N}$), then the value of $(a + k)$ is
2. If length of the common tangent of circle $x^2 + y^2 = 4$ and conic $3x^2 + 5y^2 = 15$ is
3. A circle of radius r , touches the hypotenuse of a right angled triangle at its middle point and passes through the middle point of shorter side. If 3 and 4 are lengths of perpendicular sides, then the value of r is

Space for Rough Work

4. The slope of the chord of hyperbola $2x^2 - 3y^2 = 6$ which is bisected at (5,3) is
5. In $\triangle ABC$, if A,B,C are in AP & $\frac{a \cos B + b \cos A}{c \cos A + a \cos C} = \frac{1}{\sqrt{3}}$, then the value of $\frac{3(\sin^2 A + \sin^2 B + \sin^2 C)}{(\sin 2B + \sin 2C)^2}$ is
6. Using only 0, 1, 2, 3, 4, if number of all four digit numbers where sum of digit is 12, is N, then digit at unit place of N, is

Space for Rough Work

7. If $T = \sum_{r=1}^{18} \frac{(-1)^{r-1} \cdot r \cdot {}^{18}C_r}{(r+1)}$ then $\left(\frac{1}{T} - 10\right)$ is equal to
8. If α, β, γ are the roots of $4x^3 - 16x^2 + 19x - 6 = 0$, then the value of the determinant
- $$\begin{vmatrix} \alpha - \beta - \gamma & 2\alpha & 2\alpha \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix}$$
- is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two section as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	32
I(ii)	Paragraph Based (One or more correct option(s))	4	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	16
II	Numerical Value Type (Up to second decimal place)	8	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	32

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator